

12 Separation Axioms

Definition 12.1

Let X be a topological space

- X is T_1 if singletons are closed.
- X is Hausdorff (T_2) if, for every pair of distinct points $x, y \in X$, $\exists$ disjoint neighborhoods of x, y
- X is regular (T_3) if it is T_1 and for every $x \in X$ and every closed $B \subset X$ with $x \notin B$, $\exists$ disjoint open sets containing x and B .
- X is normal if it is T_1 and for every pair of disjoint closed subsets $A, B \subset X$, $\exists$ disjoint open sets containing A, B .

Theorem 12.2

A subspace of a T_1, T_2, T_3 space is T_1, T_2, T_3

Theorem 12.3

Let $\{X_\alpha\}_{\alpha \in J}$ be a family of T_1, T_2, T_3 spaces. Then the product is T_1, T_2, T_3 .

Lemma 12.4

Let X be a top. space. Then X is regular $\iff$ for every $x \in X$ and any neighborhood U of x , there exists a neighborhood V of x s.t. $\bar{V} \subset U$.

Proof.

($\implies$) Suppose X is regular. Fix $x \in X$, neighborhood U of x .

Since $X \setminus U$ is closed, so can find disjoint open sets containing $x, X \setminus U$.

$X \setminus U \subset U$ and $x \in V$.

$X \setminus W$ is closed set containing V , so $\bar{V} \subset X \setminus U$, and since $X \setminus U \subset W$, we have $X \setminus W \subset U$, so $\bar{V} \subset U$.

($\impliedby$)

Let $x \in X, B \subset X$ s.t. $x \notin B$, assume B is closed.

Consider $U = X \setminus B$ is a neighborhood of x . Let V be a neighborhood of x s.t. $\bar{V} \subset U$.

Thus, $V, X \setminus \bar{V}$ are open sets separating x and B . ■

Theorem 12.5

If X is a metrizable space, X is normal.

Proof.

Let A, B be disjoint closed sets subsets of X .

For each $a \in A$, there exists an $\epsilon_a > 0$ s.t. $B(a, \epsilon_a) \cap B = \emptyset$.

For each $b \in B$, there exists an $\epsilon_b > 0$ s.t. $B(b, \epsilon_b) \cap A = \emptyset$.

Consider $B(a, \epsilon_a/2), B(b, \epsilon_b/2)$

If $z \in B(a, \epsilon_a/2) \cap B(b, \epsilon_b/2)$, then $d(a, b) \leq (\epsilon_a + \epsilon_b)/2 \leq \max(\epsilon_a, \epsilon_b)$. Which is a contradiction.

Thus,

$$\bigcup_{a \in A} B(a, \epsilon_a/2) \quad \bigcup_{b \in B} B(b, \epsilon_b/2)$$

are disjoint open sets containing A, B . ■

Theorem 12.6

Every compact Hausdorff spaces is normal.

Proof.

Use homework since closed means compact in compact Hausdorff. ■

Corollary 12.7

Manifolds are regular.

Proof.

Let M be a n -manifold. By definition, M is Hausdorff.

M is locally homeomorphic to $\mathbb{R}^n$, and $\mathbb{R}^n$ is locally compact, so M is locally compact.

Thus, M has a 1-point compactification. $\hat{M}$.

$\hat{M}$ is a compact Hausdorff space. Thus, $\hat{M}$ is normal. Thus, regular, and subspaces of regular spaces are regular. ■

Theorem 12.8

Every second countable regular space is normal.

Proof.

Let β be a countable basis for X .

Let A, B be disjoint closed subsets of X .

Fix $a \in A$. Then there is a neighborhood U of a that is disjoint from B .

By regularity there is a neighborhood V of a s.t. $\overline{V} \subset U$.

Thus, there is a $W \in \beta$ s.t. $a \in W \subset U$.

Then do this for any point $a \in A$, and thus we get a countable open covering of A by sets whose closures are disjoint from B .

Do the same for B .

Call them $\{U_n\}$ and $\{V_n\}$.

Then for all n define:

$$U'_n = U_n \setminus \bigcup_{i=1}^n \overline{V_n}$$

$$V'_n = V_n \setminus \bigcup_{i=1}^n \overline{U_n}$$

Thus, these union is are disjoint open sets that contain A, B . ■

Theorem 12.9: Urysohn lemma

Let X be a normal space, let A, B be closed subsets. Then there exists a continuous function. $f : X \rightarrow [0, 1]$ s.t. $f(x) = 0$ for all $x \in A$, $f(x) = 1$ for all $x \in B$.

Theorem 12.10: Urysohn Metrization Theorem

Every regular space with a countable basis is metrizable

Proof.

By Theorem 12.8 we know that this space is normal.

Let X be normal and second countable.

Claim

There exists countable collection of continuous functions $f_n : X \rightarrow [0, 1]$ s.t. for every point $x \in X$, and every neighborhood U of x . $\exists n$ s.t. $f_n(x) = 1, f_n(X \setminus U) = \{0\}$

Proof.

Let β be a countable basis.

For each pair of indices n, m s.t. $\overline{B_n} \subset B_m$, define function $f_{n,m} : X \rightarrow [0, 1]$ s.t. $f_{n,m}(\overline{B_n}) = \{1\}$, $f_{n,m}(X \setminus B_n) = \{0\}$

The $f_{n,m}$ are indexed by a subset of $\mathbb{N} \times \mathbb{N}$, so this is a countable collection.

Fix $x \in X$, neighborhood U of x , $\exists B_n$ s.t. $x \in B_n \subset U$

By regularity, $\exists$ neighborhood V of x s.t. $\overline{V} \subset B_n$.

Thus, $\exists B_m$ s.t. $x \in B_m \subset V$. Thus, $x \in B_m$ and $\overline{B_m} \subset B_n$.

Thus, $f_{n,m}(x) = 1$, $f_{n,m}(X \setminus U) \subset f_{n,m}(X \setminus B_m) = \{0\}$.

Reindex to the collection $\{f_n\}$ ■

Define $F : X \rightarrow \mathbb{R}^\omega$ s.t. $F(x) = (f_1(x), f_2(x), f_3(x), \dots)$

F is continuous since each component is continuous.

Injective:

Let $x, y \in X$ s.t. $x \neq y$. X is Hausdorff, so there exists a neighborhood U of x s.t. $y \notin U$.

Thus, there exists n s.t. $f_n(x) > 0$ and $f_n(z) = 0$ for all $z \notin U$, so $f_n(y) = 0 \implies f_n(x) \neq f_n(y) \implies F(x) \neq F(y)$.

Let $Z = F(X)$.

Let U be open in X . Consider $F(U)$.

Let $z \in F(U)$ thus there exists $x \in U$ s.t. $F(x) = z$. Thus, there is a n s.t. $f_n(x) > 0$ and $f_n(X \setminus U) = \{0\}$.

Consider the index- n projection map $\pi_n : \mathbb{R}^\omega \rightarrow \mathbb{R}$.

Consider the set $V = \pi_n^{-1}((0, \infty)) \subset \mathbb{R}^\omega$.

$f_n(x) > 0 \implies f(x) = z \in V$.

Let $y \in V \cap Z$ then $y = F(c)$ for some $c \in X$, so $f_n(c) > 0$. But then $c \in U$, since $f_n(X \setminus U) = \{0\}$.

Thus, $V \cap Z \subset F(U)$, so $F(U)$ is open. Thus, F is an embedding into $\mathbb{R}^\omega$. Thus, X is homeomorphic to a subspace of a metric space, so X is a metric space. ■